Paleomagnetism of the komatiitic basalts of the Ottawa Islands, N.W.T.

K. L. Buchan and W. R. A. Baragar
Geological Survey of Canada, 601 Booth St., Ottawa, Ont., Canada K1A 0E8

Received June 12, 1984
Revision accepted October 30, 1984

The komatiitic basalts of the Ottawa Islands in eastern Hudson Bay are on strike with and believed to form a continuation of similar units of the Cape Smith Belt 150 km to the northeast. Units sampled in the Ottawa Islands all dip gently to the west and hence are not suitable for an internal fold test of their age of magnetization. However, before correcting for the tilt of the lavas, the dominant magnetization direction (D = 207.6°, I = 61.9°, k = 168, α_{95} = 3.7°) does not differ significantly from the uncorrected magnetization direction reported from the steeply dipping, northwest-facing units at Cape Smith (D = 218°, I = 60°, k = 47, α_{95} = 4°). This negative fold test suggests that the remanence at both locations was acquired after folding. Comparison with the North American Precambrian apparent polar wander path implies that overprinting is related to the Hudsonian Orogeny.

A second stable magnetization directed to the west with a shallow inclination is superimposed on the dominant component at a number of sampling sites. Its direction is poorly defined and no fold test is possible. However, magnetic evidence suggests that this component was probably acquired as an overprint after the dominant magnetization, perhaps during a mild reheating associated with the Elsonian Orogeny.

Les basaltes komatiitiques des îles Ottawa de l'est de la baie d'Hudson sont de même direction et semblent former une continuation avec les unités analogues de la ceinture de Cape Smith à 150 km au nord-est. Les unités échantillonnées dans les îles Ottawa présentent toutes un faible pendage vers l'ouest et par conséquent le test de plissement interne pour leur âge d'aimantation ne peut être appliqué adéquatement. Cependant, avant correction pour l'inclinaison des laves, la direction d'aimantation dominante (D = 207,6°, I = 61,9°, k = 168, α_{95} = 3,7°) ne diffère pas significativement de la direction d'aimantation non corrigée obtenue pour les pendages abrupts des unités faisant face au nord-ouest à Cape Smith (D = 218°, I = 60°, k = 47, α_{95} = 4°). Ce test de plissement négatif indique que la rémanence aux deux emplacements fut acquise antérieurement à la phase de plissement. Une comparaison avec la courbe de la dérive apparente des pôles du Précambrien de l'Amérique du Nord implique qu'une surimpression est associée à l'orogenèse de l'Hudsonien.

Une deuxième aimantation stable de direction ouest caractérisée par une inclinaison peu profonde se superpose à la composante dominante à un certain nombre de sites d'échantillonnage. Sa direction est mal définie et l'application du test de plissement est impossible. Cependant, les mesures magnétiques suggèrent que cette composante aurait pu être acquise sous forme d'une surimpression ultérieurement à l'acquisition de l'aimantation dominante, possiblement durant un réchauffement modéré associé avec l'orogenèse de l'Elsonien.

[Traduit par le journal]

Can. J. Earth Sci. 22, 553–566 (1985)

Introduction

The komatiitic basalts of the Ottawa Islands are thought (Baragar and Piché 1982) to be an extension of the komatiitic suite of the Cape Smith Belt of northern Quebec.

At Cape Smith, the steeply dipping, northwest-facing komatiitic lavas are uniformly magnetized (Fujiwara and Schwarz 1975) with a remanence direction of D = 218°, I = 60°, k = 47, α_{95} = 4°. Since they dip uniformly, their magnetization neither clearly pre- nor postdates folding. Farther south, Schmidt (1980) has shown that a similar magnetization in the Belcher Islands was acquired after folding. This suggests that the Cape Smith remanence may also be a secondary overprint.

To test this hypothesis, samples were measured from 21 sites on four of the Ottawa Islands, where, in contrast to the steeply dipping beds at Cape Smith, the komatiitic lavas dip gently to moderately (15–40°) westward.

Geological setting

In eastern Hudson Bay and northern Quebec the late Aphebian Circum-Superior Belt (Fig. 1) is formed predominantly of volcanic rocks comprising two distinctive stratigraphic and petrochemical groupings. In the Cape Smith segment of the belt the older grouping is termed the Povungnituk and the younger the Chukotat Group (Bergeron 1959). They were thought to be separated by an unconformity but later work (Taylor 1974, 1982; Baragar 1974) was not able to confirm this and it may be in doubt. More recently the volcanic rocks composing the Povungnituk and Chukotat groups have been found to have affinities to tholeiitic and komatiitic magma types, respectively (Schwarz and Fujiwara 1977; Francis and Hynes 1979).

Strata of the Circum-Superior Belt rest unconformably on Archean basement and dip away from it north and westward at varying angles. The tholeiitic rocks are generally on the inner side of the belt closest to the Archean basement and are found along the southern side of the Cape Smith Belt and on the Sleeper and Belcher islands. Rocks of the komatiitic suite form the central rib of the Cape Smith Belt and extend into the Ottawa Islands. Metamorphism increases away from the stable Archean core and in the Cape Smith region, at least, passes northward into high-grade schists and gneisses. The deformation is Hudsonian but K–Ar dates on metamorphic rocks and minerals range from about 1850 to 1430 Ma (Beall *et al.* 1963). The age of the volcanic rocks is also poorly constrained. Dates of 2150 Ma (Schimann 1978; Rb–Sr) and 1870 Ma (Zindler 1981; Sm–Nd) were obtained on volcanic rocks of the Cape Smith Belt, but on rocks that might be judged to be roughly equivalent in the Belcher Islands, Fryer (1972) obtained an Rb–Sr date of 1705 Ma and Brooks and Arndt (quoted in Schmidt 1980) obtained an Rb–Sr date of 1590 Ma. If we assume the Hudsonian Orogeny peaked at about 1800 Ma (Stockwell 1982), a reasonable age for rocks of the Circum-Superior Belt might be 1900–2000 Ma.

In the Ottawa Islands volcanic rocks of the komatiitic suite (Baragar and Lamontagne 1980; Baragar and Piché 1982) were

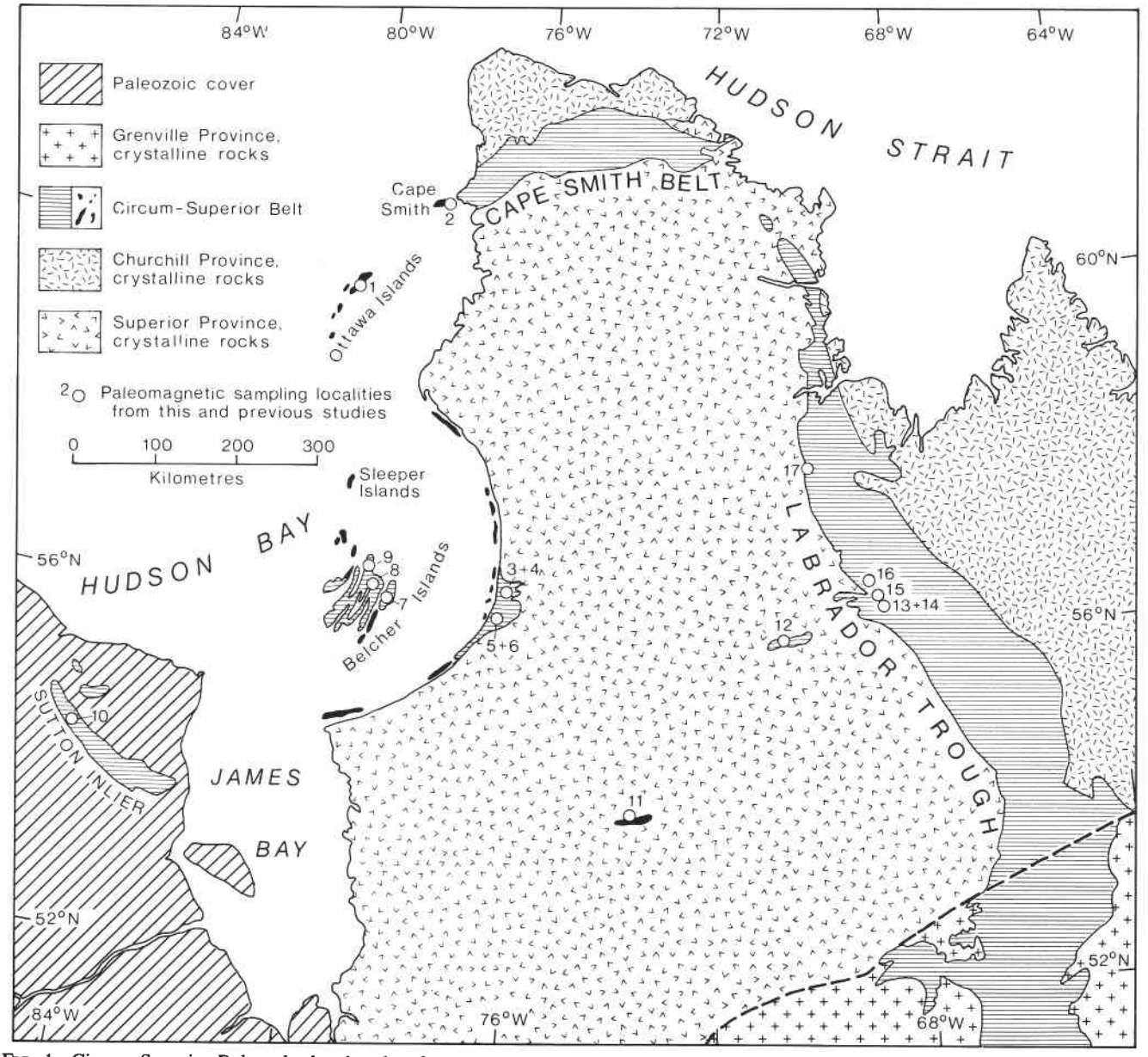


FIG. 1. Circum-Superior Belt and related rocks of eastern Hudson Bay and Ungava, Canada. Localities of paleomagnetic studies identified on the map are (1) Ottawa Islands volcanics; (2) Cape Smith volcanics; (3) Quingaaluk sediments; (4) Persillon Formation (north); (5) Persillon Formation (south); (6) Nastapoka basalts; (7) Haig intrusions; (8) Eskimo volcanics; (9) Flaherty volcanics; (10 Sutton Lake inlier; (11) Sakami Formation; (12) Dieter Lake redbeds; (13) Chakonipau Formation; (14) Seward Subgroup; (15) Castignon Complex; (16) Menihek Formation; (17) Lac Imbault redbeds. Studies (2)–(17) are reported in Fujiwara and Schwarz (1975), Park (1977), Schmidt (1980), Schwarz and Fujiwara (1981), Seguin *et al.* (1981), Schwarz *et al.* (1982), and Schwarz and Freda (1983).

subdivided into low-, medium-, and high-Mg varieties on the basis of their physical characteristics and some representative chemical analyses. In general the divisions are at about 10 and 16% MgO, respectively, but because the subdivision is field based these can only be regarded as approximations (Baragar 1983). By the definition agreed upon at the Penrose Conference on Komatiites (Arndt and Brooks 1980) these rocks would be classed as basaltic komatiites, komatiitic basalts, and basalts. However, their generally high MgO and low TiO_2 contents would suggest that they all belong to a consanguineous suite referred to here as the komatiitic suite. Pillowed and massive lavas are finely interlayered throughout the succession, which would appear to indicate that their eruption was entirely subaqueous. Some of the massive lavas are fractionated and so comprise an olivine-cumulate layer near their base and variously fractionated basalts in their upper levels. More rarely the upper part of such layered flows is marked by coarse spinifex-textured pyroxene and olivine.

The volcanic succession sampled for the paleomagnetic studies (Fig. 2) forms a homoclinal sequence facing northwest and ranging in dip from about 40 to 45° westerly on the eastern side of Gilmour Island to 15° on the western side. Repetition of parts of the sequence by faulting is possible, since lineaments can be seen on air photographs, but no direct evidence for this is available. Assuming an unfaulted sequence, the thickness represented in the map of Fig. 2 would be about 4500 m.

Sampling and laboratory procedures

Twenty-one paleomagnetic sampling sites (numbered 25–45 in Fig. 2) are distributed fairly evenly across the ex-

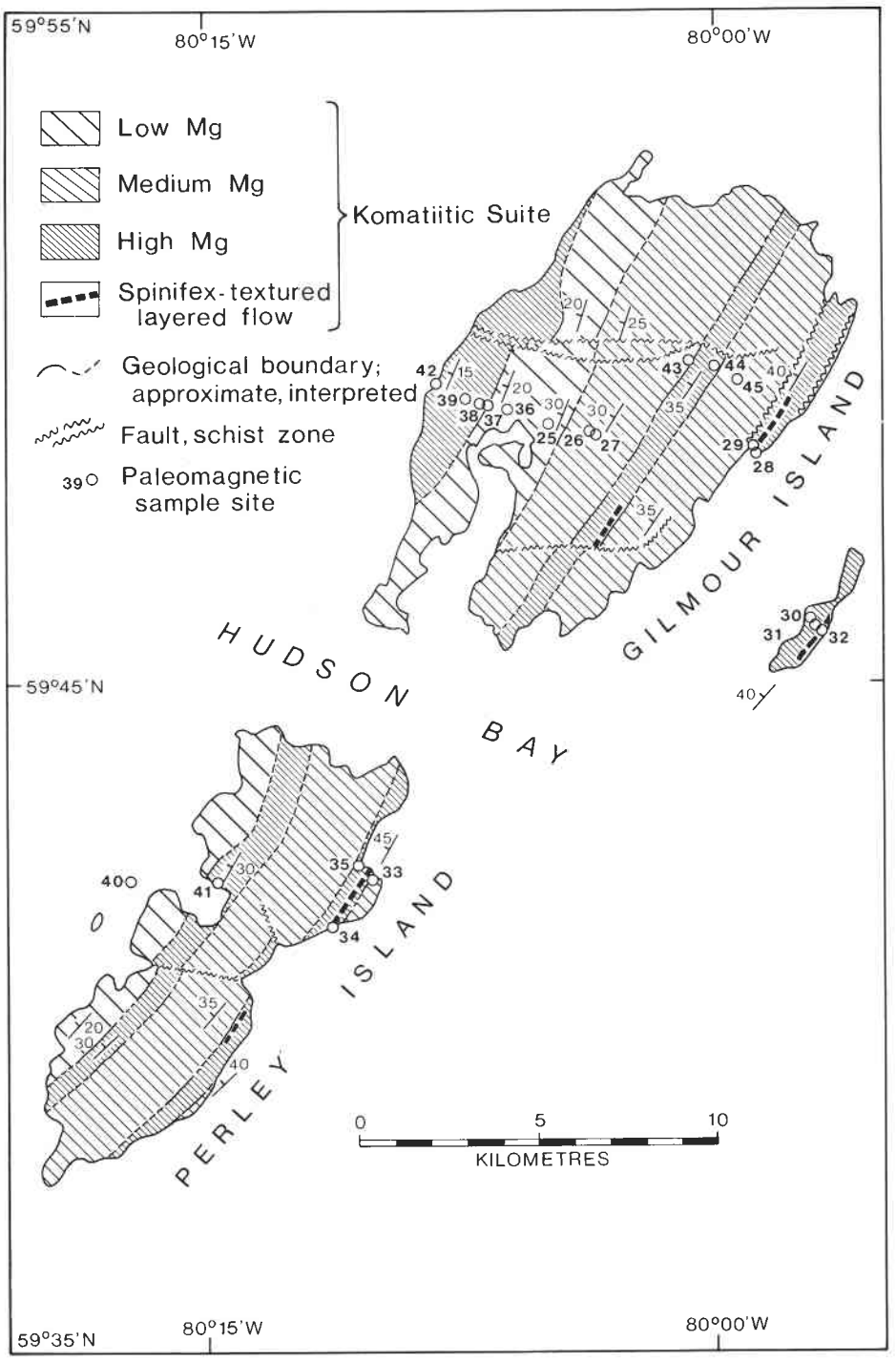


FIG. 2. Geology and paleomagnetic sites of the Ottawa Islands.

posed thickness of the sequence and represent all major rock types including the normal and layered spinifex-textured flows. About five drill cores were oriented with sun and magnetic compasses at each site. Two or more specimens (2.5 cm in diameter and 2.2 cm in length) were cut from each core.

Usually, two specimens per core were stepwise alternating field (AF) demagnetized to peak fields of 0, 2.5, 5.0, 10, 20, 25, 30, 35, 40, 50, 60, 70, 80, 90, and 100 mT (1.0 mT = 10 Oe) using a commercial AF demagnetizer. Specimen magnetizations were measured on a commercial spinner magnetometer. Second specimens from unstably magnetized cores were not demagnetized. During stepwise demagnetization a specimen's magnetization was used for directional analysis until its intensity fell below 5×10^{-5} A/m (1 A/m = 10^{-3} emu/cm^3), when instrumental noise became a serious problem.

Twenty-five specimens from 17 sites were thermally cleaned at peak temperatures of 20, 100, 175, 250, 300, 350, 400, 450, 500, 530, 550, 560, 570, 580, 590, and 600°C in order to investigate the stability of observed magnetization components upon thermal demagnetization and to determine their relative (and absolute) blocking temperatures. A commercial thermal demagnetizer was employed.

Occasionally, specimens were AF demagnetized to peak fields of 10 or 15 mT before the stepwise thermal demagnetiza-

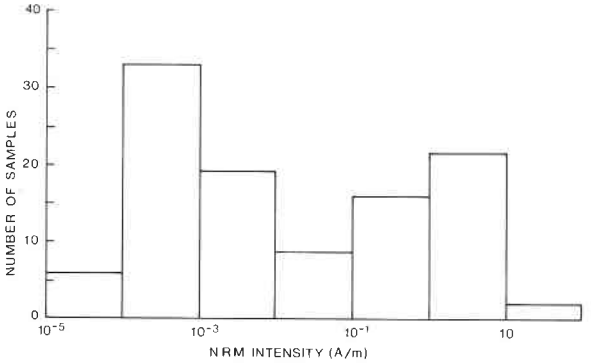


FIG. 3. Histogram of NRM intensities.

tion was carried out. This effectively eliminated viscous components and allowed a more detailed examination of underlying stable components.

Results

Natural remanent magnetization (NRM) directions are rather scattered, with a predominance of steep positive inclinations. NRM intensities, which range between 10^{-5} and 10^{2} A/m, display a bimodal distribution (Fig. 3) with peaks three or four orders of magnitude apart.

In general, high-intensity specimens (NRM $> 10^{-1}$ A/m) attain stable magnetization directions on AF and thermal demagnetization. Low-intensity specimens (NRM $< 10^{-3}$ A/m) and intermediate-intensity specimens have a much broader range of stability. Both high- and low-intensity specimens, however, show evidence of two stable magnetization directions, a dominant A component directed to the south-southwest with a steep positive inclination (similar to the Cape Smith magnetization) and a B magnetization directed to the west with a shallow inclination.

AF demagnetization

On AF demagnetization, most high-intensity specimens reach a stable A end point in a south-southwest direction with a steep, positive inclination (Fig. 4*a*). Typically the A end point is attained in peak AF's of 15 mT or less and maintained to fields of 50 mT or more. On a corresponding Zijderveld diagram of the horizontal and vertical components of magnetization (Fig. 4*a*), a straight line segment directed towards the origin is observed after elimination of the viscous component in 15 mT. Occasionally, low- (or intermediate-) intensity specimens also stabilize in the same direction (Fig. 4*b*) although scatter of data points is often greater than that observed with high-intensity specimens.

Other stable specimens of the Ottawa Islands collection show evidence of a second component with a shallow, westerly direction. However, this B magnetization is usually superimposed on the A component. Thus the resultant directions fall along a rough great circle between the A and B end points.

For example, at high-intensity site 26 (Fig. 5 *a* and *b*) a stable B direction is isolated in a specimen from core 2601 and a stable A direction in a specimen from core 2604. The corresponding Zijderveld diagram for core 2601 is shown in Fig. 5*c*. At low-intensity site 33 (Fig. 5*d*) individual specimen magnetizations stabilize along the great circle defined by the A and B directions. In other (usually low-intensity) cases only stable intermediate directions are observed (e.g., 4004 of Fig. 5 *e* and *f*). A viscous component is eliminated in 15 or

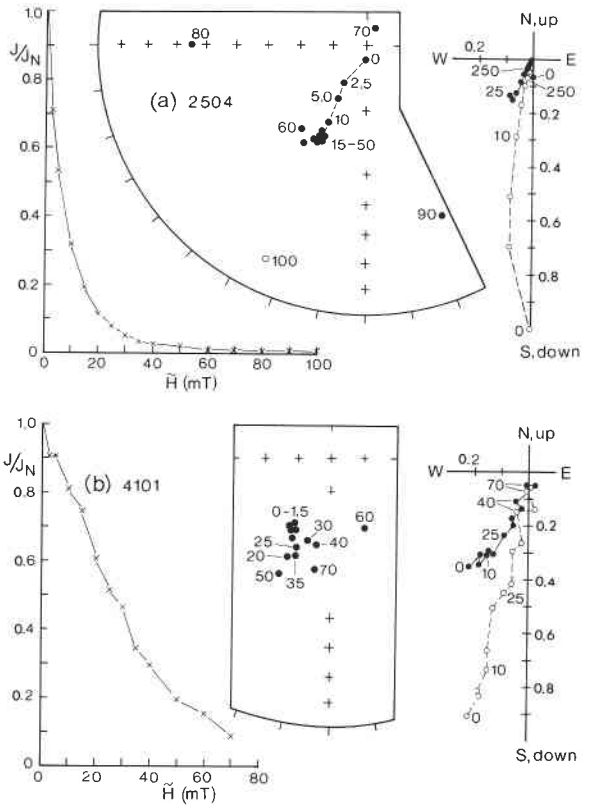


FIG. 4. Typical A directions and intensities obtained by AF demagnetization of specimens from (*a*) a high-intensity core and (*b*) a low-intensity core. Intensities are normalized to the NRM value. Directions are plotted on an equal-area projection with open (closed) symbols indicating upward (downward) vectors. Zijderveld diagrams show the projection of magnetization vectors onto the horizontal plane (closed symbols) and east–west vertical plane (open symbols). Axes are normalized to the total NRM intensity. Alternating fields are in milliteslas.

20 mT, then the data points of the Zijderveld diagram follow roughly linear paths towards the origin, becoming increasingly scattered above 40 mT as the instrumental noise level is approached. Interestingly, there is no evidence on either the stereoplot (Fig. 5*e*) or the Zijderveld diagram (Fig. 5*f*) of the two-component nature of the remanence above 15 mT. However, a second specimen from the same core (not shown) does show a short swing along the A–B great circle.

Perhaps one of the most dramatic cases of superposition occurs at site 32 (Fig. 6). Here the NRM of each specimen is directed steeply down, suggesting a large viscous component parallel to the Earth's magnetic field. Upon stepwise AF demagnetization the direction becomes progressively shallower as the viscous component is erased (0–15 mT) and then follows a great circle path towards the A end point as the B component is eliminated (15–50 mT).

When horizontal and vertical components of magnetization at site 32 are plotted on a Zijderveld diagram (e.g., sample 3204 in Fig. 6), the data points do not define straight line segments until only the A component remains above 50 mT. Apparently both A and B components are erased simultaneously, but at different rates, between 15 and 50 mT. Thus the Zijderveld diagrams cannot be used to define the B direction.

At several (usually low-intensity) sites, individual cores have stable A, B, or intermediate directions, whereas others are unstably magnetized or too weakly magnetized (intensity <

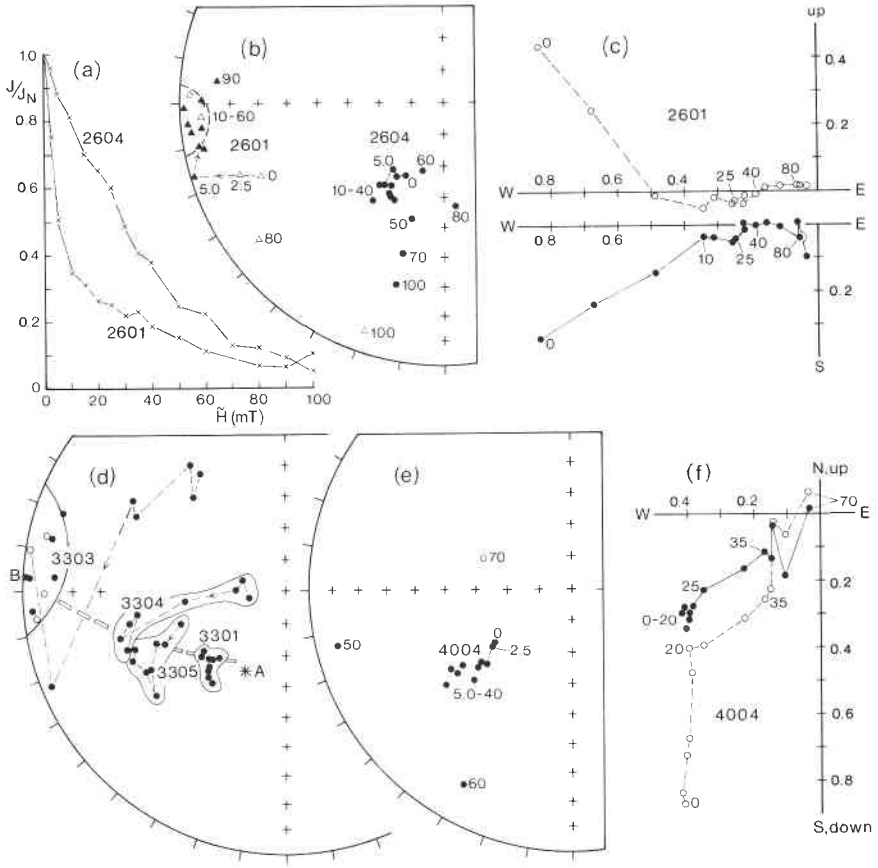


FIG. 5. AF demagnetization of specimens from superimposed A−B sites. Symbols are the same as in Fig. 4. (*a*), (*b*), (*c*) Stable A and B results from a single site. (*d*) Directions from a single site smeared along the great circle between A and B. (*e*), (*f*) Example of a stable direction intermediate between A and B. Alternating fields are in milliteslas.

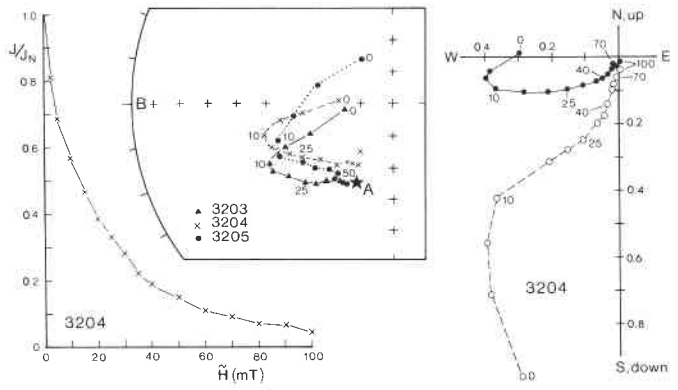


FIG. 6. AF demagnetization of typical specimens at site 32. Directions move systematically away from the present Earth's field between 0 and 15 mT, then follow the A−B great circle towards the A end point. Directions above 50 mT are not shown to avoid congestion. Symbols are the same as in Fig. 4. Alternating fields are in milliteslas.

5×10^{-5} A/m) to be measured accurately with a commercial spinner magnetometer. At two sites no cores were stable during AF cleaning.

Thermal demagnetization

In an attempt to better establish the B direction and in order to investigate the relative blocking temperatures of A and B magnetizations, 25 representative specimens were selected from 17 sampling sites and were thermally demagnetized in stepwise fashion. In several cores, pilot thermal demagnetization experiments demonstrated that viscous components are large enough and have a sufficiently broad range of blocking temperatures that they effectively obscure details of the smaller A and (or) B components. Such specimens were AF demagnetized to 15 mT (or less) to erase the viscous component before thermal demagnetization was begun.

Both high- and low-intensity cores that on AF demagnetization have a stable A direction also maintain a stable A direction upon thermal demagnetization (Fig. 7). Blocking temperatures of high-intensity samples are consistent with magnetite as the dominant carrier of stable magnetization (Fig. 7*a*). Blocking temperatures of low-intensity samples are more variable. They may form a broad range below 580°C or be concentrated over narrow ranges at lower temperatures (Fig. 7*b*).

Samples that on AF demagnetization show evidence of a B component often behave quite differently on thermal cleaning. For example, on AF demagnetization the magnetization of a specimen from core 2701 swings slowly from the B direction to the A direction, stabilizing in about 30 mT (Fig. 8*a*). However, on thermal demagnetization (Fig. 8*a*) a stable B direction is maintained throughout the demagnetization of another specimen from the same core. In another case, one specimen of sample 3505 moves to a B direction on AF demagnetization, whereas a second swings to the A end point on thermal demagnetization (Fig. 8*b*). In the latter case, the large increase in intensity above 530°C after the directions have randomized is unexplained but probably indicates serious chemical alteration of the specimens during heating. Such chemical effects do not

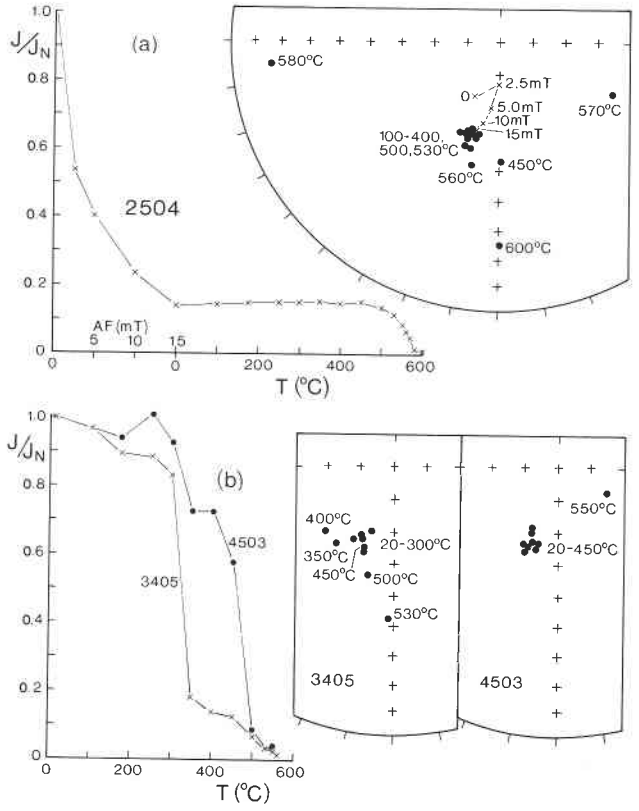


FIG. 7. Thermal (or combined AF−thermal) demagnetization of (*a*) a typical high-intensity A specimen and (*b*) typical low-intensity A specimens. Symbols are the same as in Fig. 4.

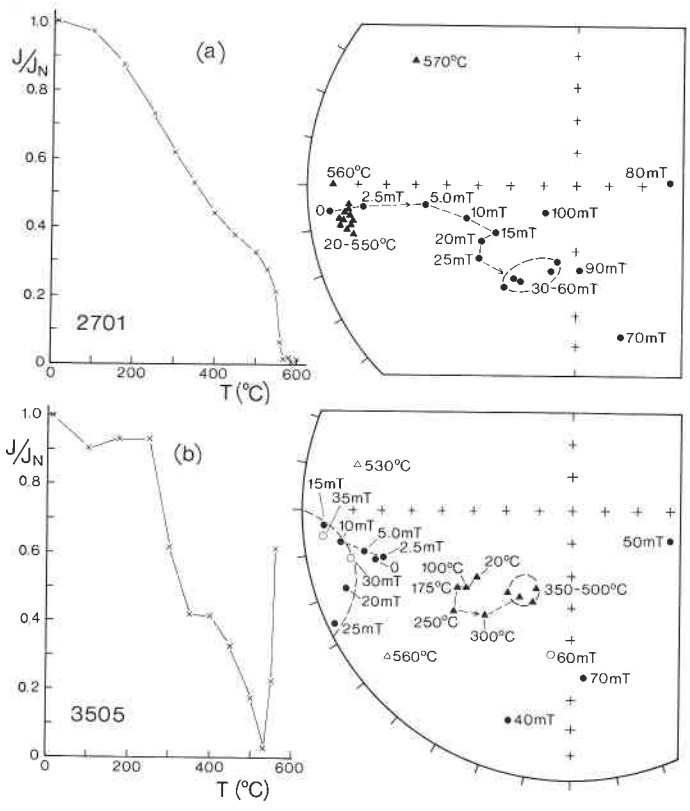


FIG. 8. Comparison of thermal and AF demagnetization of specimens from the same core at superimposed A−B sites. Symbols are the same as in Fig. 4.

occur elsewhere in the Ottawa Islands thermal results.

Site 32 provides a particularly striking example of the difference in behaviour that is sometimes observed between AF and thermal demagnetization (Fig. 9). On AF demagnetization (recall Fig. 6) directions consistently swing from midway along the A−B great-circle arc to the A direction after elimination of a viscous component. However, on thermal cleaning (Fig. 9) a viscous component is eliminated in AF's of 10 or 15 mT; the direction then remains stable midway along the A−B great circle to about 500°C (since there are no blocking temperatures in this range) before swinging rapidly towards the B end point. Unfortunately, the directions randomize at 560 or 580°C, so a stable B end point is never achieved.

Occasionally, A−B samples exhibit identical intermediate directions upon AF and thermal cleaning. An example is shown in Fig. 10.

B blocking temperatures are quite variable. They may be concentrated close to the Curie temperature of magnetite (e.g., Fig. 8*a*) or form a broad temperature range (e.g., Fig. 10). A and B blocking temperatures do not, however, occupy exclusive temperature ranges. In some mixed A−B specimens (e.g., Fig. 8*b*) the A blocking temperatures are above B blocking temperatures; in others (e.g., Fig. 9) the reverse is true. In many cases (e.g., Fig. 10) the blocking temperatures overlap.

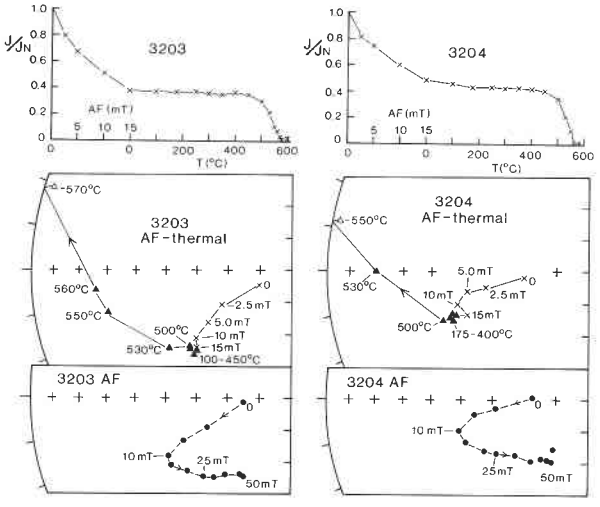


FIG. 9. Comparison of combined AF−thermal demagnetization and AF demagnetization of specimens from the same cores at site 32. Symbols are the same as in Fig. 4.

Analysis

At 12 sampling sites, magnetization directions reach consistent end points on AF demagnetization. Core mean directions, based in each case on the AF demagnetization step (≥15 mT) with the lowest value of θ (the angular separation between the directions of the two specimens), are plotted in Fig. 11*a*. Site means are shown in Fig. 11*b* and listed in Table 1. Two of the site means (sites 40 and 42) are displaced significantly from the main grouping toward the shallow, westerly B direction. As these sites likely contain superimposed A and B components, they have not been incorporated in the A mean direction of $D = 207.6°$, $I = 61.9°$ ($k = 168$, $\alpha_{95} = 3.7°$). The corresponding A paleopole falls at 100.6°W, 15.4°N (dm = 5.8°, dp = 4.5°). It should be noted that inclusion of sites 40 and

TABLE 1. Mean directions at all stable sites (before unfolding)

Site	Int.	N	n	AF (mT)	D (°)	I (°)	k	α_{95} (°)
				A sites				
25	H to I	6	6	15–40	207.7	59.6	133	5.8
27[a]	H	6	4	15–40	223.4	65.6	18	22.3
30	H	5	5	15–35	192.5	65.7	577	3.2
31	H	5	5	15–60	195.3	58.8	222	5.2
32	H	5	5	15–90	214.8	63.2	164	6.0
34	H to L	5	5	15–25	197.6	57.8	41	12.0
39	H	5	5	15–25	227.5	64.4	26	15.2
41	I to L	5	2	15–50	205.1	61.9	2263	5.3
44	H	5	5	15–90	203.6	61.3	210	5.3
45	L	4	4	15–25	212.4	56.5	26	18.4
A core mean			46	—	207.2	61.9	51	3.0
A site mean			10	—	207.6	61.9	168	3.7
Mean A paleopole[b]					100.6W	15.4N	dm=5.8°	dp=4.5°
				Superimposed A–B sites				
40[c]	L	6	2	15–35	235.8	46.6	382	12.8
42[c]	L to I	6	6	15–50	239.0	57.9	97	6.8

NOTES: Int. is the average NRM intensity (H = high; I = intermediate; L = low); N is the number of cores measured (two specimens per core); n is the number of cores used in the statistical calculations (only sites for which $n \geq 2$ are included); AF is the range of peak alternating fields used to calculate the mean direction; k is the precision parameter; and α_{95} is radius of the circle of 95% confidence.

[a]One core (2701), which gives a B end point, is included in Table 2.

[b]The mean paleopole is calculated from the site mean directions.

[c]Individual core statistics for sites 40 and 42 are listed in Table 2.

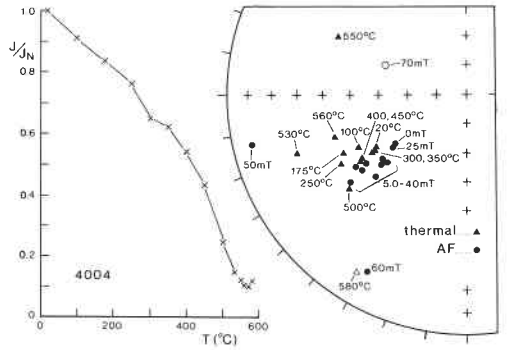


FIG. 10. Comparison of thermal and AF demagnetization characteristics of a core that yields a stable A–B direction. Symbols are the same as in Fig. 4.

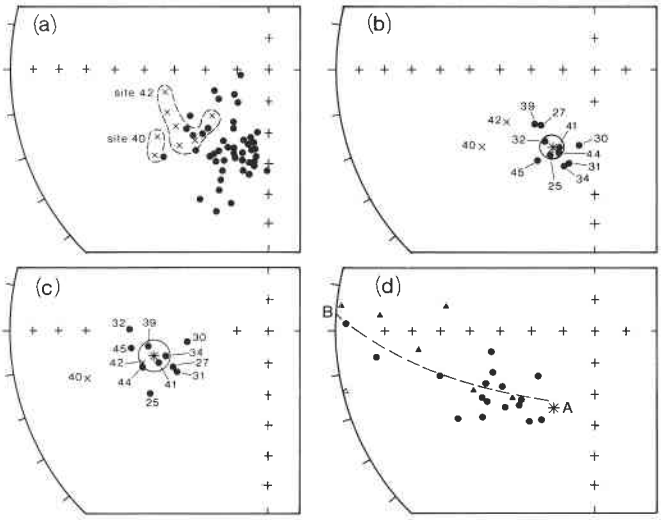


FIG. 11. Core and site mean directions of magnetization after optimum AF cleaning. (*a*) Core mean directions before correction for tilt of bedding for sites where directions reach stable, consistent end points. Directions from two sites (crosses) are thought to contain superimposed A and B components. (*b*) Corresponding site mean directions before bedding correction. The A mean direction is shown as a star with its circle of 95% confidence. (*c*) Corresponding site mean directions after bedding correction. (*d*) Stable core (circles) or specimen (triangles) end points from sites with superimposed A and B components. "Swinging" samples of site 32 are not included in (*d*).

42 in the calculation of the A mean (D = 213.6°, I = 60.8°, k = 73, α_{95} = 5.1°) would not alter the interpretation discussed below.

When the site mean directions of Fig. 11*b* are corrected for the tilt of the bedding (Fig. 11*c*), the mean direction (of the 10 A sites) moves to D = 257.2°, I = 53.0° and the scatter increases slightly (k = 98.4, α_{95} = 4.9). However, a comparison of uncorrected and tilt-corrected results for the Ottawa Islands is inconclusive because the beds all dip rather uniformly (15–40° to the west).

A more convincing fold test is obtained by comparing the paleomagnetic pole from the Ottawa Islands lavas with their counterparts in the Cape Smith region 150 km to the northeast. At Cape Smith, the komatiitic basalt flows are tilted nearly vertically, yet their mean paleopole 107°W, 16°N, dm = 5°, dp = 4° (Fujiwara and Schwarz 1975), uncorrected for the tilt of bedding, is not significantly different at the 95% confidence level from the uncorrected A paleopole observed on the Ottawa Islands. After unfolding, however, the two poles differ by ~52°. Thus, this constitutes a negative fold test for the komatiitic basalts of Cape Smith and the Ottawa Islands and indicates that the A magnetization at both locations is postfolding.

In addition to sites 40 and 42, which have consistent, stable directions of magnetization offset from the main A grouping (Fig. 11*b*), there are seven other sites at which directions of

TABLE 2. Directions (before unfolding) for stable[a] cores from superimposed A−B sites

Site	Core	Int.	N	n	AF/th	D (°)	I (°)	θ (°)
26	01	I	2	2	40 mT	271.5	4.6	6.5
	04	I	2	2	20 mT	211.2	57.2	1.0
27	01	H	2	2	550°C	263.1	16.1	3.2
28	02	L	1	1	15−35 mT	279.6	41.9	—
33	01	L	2	2	30 mT	216.2	54.2	3.7
	03	L	2	1	25−100 mT	275.5	1.9	—
	04	L	2	2	20 mT	253.5	37.9	8.0
	05	L	2	2	20 mT	236.7	48.9	20.0
35	01	L	2	2	20 mT	244.1	51.1	7.2
	03	L	2	1	15−30 mT	263.9	32.2	—
	04	I	2	2	30 mT	225.7	56.5	2.5
	05	L	2	1	15−35 mT	256.5	−0.5	—
36[b]	04	L	1	1	15−35 mT	244.3	46.9	—
	04	L	1	1	15−40 mT	274.3	18.2	—
37	01	L	2	1	20 mT	230.5	66.6	—
	02	I	2	2	15 mT	237.6	37.5	7.4
40[c]	03	L	2	2	35 mT	239.0	48.6	8.1
	04	L	2	2	15 mT	232.9	44.5	2.6
42[c]	01	L	2	2	15 mT	258.1	56.3	2.2
	02	L	2	2	25 mT	229.5	52.7	12.6
	03	I	2	2	20 mT	239.0	55.7	1.5
	04	I	2	2	15 mT	247.0	55.0	2.2
	05	I	2	2	20 mT	226.7	57.8	2.2
	06	I	2	2	50 mT	231.1	67.0	2.1

NOTE: Int is the NRM intensity (H = high; I = intermediate; L = low); N is the number of specimens measured; n is the number of specimens used in the statistical calculations; AF/th is the peak alternating field(s) or peak temperature(s) used to calculate the mean direction; θ is the angle between specimen directions.

[a]Samples that swing along the A−B great circle (see Fig. 6) are not included in this table.

[b]Individual specimens from core 3604 give distinctly different stable directions along the A−B great circle.

[c]Mean directions for sites 40 and 42 are listed in Table 1.

one or more individual cores or specimens are scattered along rough great circles between the A and B end points (see, for example, Fig. 5*d*). Means for individual stable cores (or specimens) at all such sites are plotted in Fig. 11*d* and listed in Table 2. The B direction must, of course, lie at or beyond the end of the great-circle arc defined by these directions.

As noted earlier, all specimens at site 32 exhibit swings along the A−B great circle (see Figs. 6, 9). In an attempt to estimate the direction of the B component in such samples, a modified version (Halls 1979) of Hoffman and Day's (1978) method for isolating the direction of the intermediate-coercivity component in specimens that carry three superimposed magnetizations was applied to the data of site 32. In the case of AF demagnetization the three components involved are the soft viscous magnetization, the B component of intermediate stability, and the harder A magnetization.

In the Hoffman−Day (1978) method the component erased after each successive AF demagnetization is plotted. A great circle is drawn through points in the range where only the two softest components are eliminated; a second is drawn through points where only the two hardest components are erased. The direction of the intermediate-stability component is then defined by the intersection of these two great circles. In Halls' (1979) modified version of the method, the *latter* great circle is replaced by one that is defined by the measured directions over the same range of coercivities (or temperatures), since these "resultant" vectors show much less scatter than the corresponding difference vectors.

The modified Hoffman−Day technique of Halls (1979) was applied to the AF demagnetization data of site 32 (Fig. 12 *a* and *b*). Only about one third of the specimens yield useful results. As expected, the great-circle intersections define a shallow, westerly direction. In the remaining specimens the scatter of difference vectors is such as to preclude the definition of great circles.

In Fig. 12*c* the above technique is modified slightly in order to examine the combined AF−thermal demagnetization results for a single specimen from site 32. Vector differences during AF cleaning to 10 or 15 mT define a great circle, as the soft viscous component is eliminated more quickly than B. When demagnetization is continued thermally above 15 mT, however, the B component can no longer be considered to have stability intermediate between that of the VRM and A, since resultant vectors swing toward the B, not the A, direction. Nevertheless, the great circle defined by this latter swing is just as useful as in the case of AF demagnetization to 100 mT. Once again the two great circles intersect to define a shallow, westerly B direction.

A total of six specimens at site 32 give usable B intersections with the Hoffman−Day technique (Fig. 12*d*). They support a shallow, westerly B direction.

Because of the paucity of B data from swinging and stable sites, the uncertainty about whether the data are sufficient to average out secular variation, and the difficulties in separating the A and B components in superimposed samples, no mean B direction is reported. Only a rough estimate of the location of the B paleopole is used in the discussion below. Nevertheless, the evidence clearly supports a B component with a shallow, westerly direction.

Although stable magnetizations intermediate between A and B and magnetizations that swing between A and B have been treated in this paper as the superposition of distinct A and B components, it should be noted that there are other possible interpretations. Swinging directions could record apparent polar wander (see, for example, Stearn and Piper 1984), whereas stable intermediate remanences might be either thermoremanent magnetization acquired at some time between the acquisition of A and B or chemical remanent magnetization acquired at the time of B in the presence of the earlier A component (Bailey and Hale 1981).

In the present study it seems unlikely that swinging or intermediate magnetizations record apparent polar wander between the time of A and B, since intermediate magnetizations and swings fall along a rough great circle (Figs. 5, 6, 11) while the polar wander path for North America for the postulated period between A and B (see Discussion) is undoubtedly much more complex. In addition, the Hoffman−Day analysis of swinging samples at site 32 (Fig. 12) would not yield great circles if apparent polar wander were recorded by the swings. On the other hand, it is not possible to discount the possibility that stable intermediate directions were acquired as chemical remanent magnetization at the time of B in the presence of an earlier A component.

Petrography

Polished thin sections examined from each of the paleomagnetic sites (Table 3) provide a fairly representative sam-

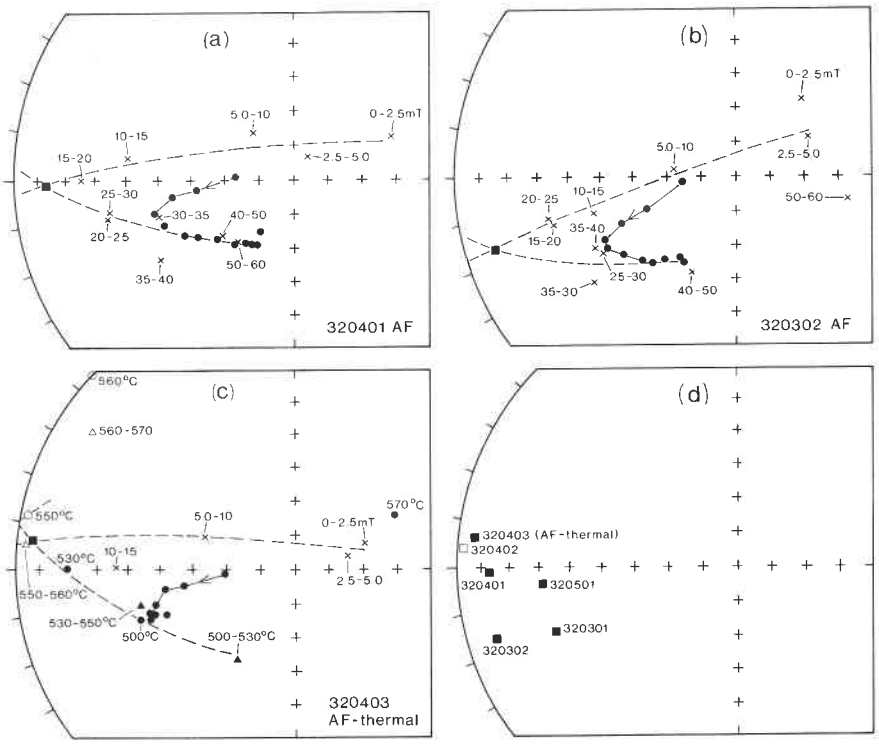


FIG. 12. Estimation of the B direction at site 32 using the modified Hoffman–Day technique of Halls (1979). (*a*) AF results for specimen 320401; (*b*) AF results for specimen 320302; (*c*) AF–thermal results for specimen 320403; and (*d*) B directions estimated from modified Hoffman–Day techniques for site 32. Circles (triangles) are the AF (thermal) demagnetization data points. Crosses are the corresponding directions of subtracted vectors. Great circles are determined as described in the text. Squares indicate the intersection of great circles.

pling of the petrography of the islands shown in Fig. 2. The rocks are generally metamorphosed to greenschist facies but commonly contain relicts of the primary minerals in varying proportions. Deformation is slight or absent and most rocks retain their igneous texture sufficiently well that their primary mineralogy can be readily interpreted.

The rocks were originally composed of olivine and clinopyroxene mainly, with accessory chromite, magnetite, and, to a lesser degree, chalcopyrite and pyrite. Plagioclase is present in the low-Mg lavas but is not generally in excess of about 10 or 15%. Olivine typically occurred as phenocrysts composing from about 15 to about 50% of the rock and was rarely present in the groundmass. Clinopyroxene, on the other hand, was a subordinate phenocryst phase but formed the bulk of the groundmass where it generally appeared as parallel or plumose acicular crystals commonly interleaved with plagioclase or opaque-charged glass (now chlorite). Chromite most commonly crystallized in or adjoining olivine phenocrysts. Although in most samples magnetite appears to have been a late crystallizing phase that occupied the interstices of the groundmass, in places it formed coarser crystals that were at least as early in the crystallization sequence as the groundmass clinopyroxene.

Metamorphism has been somewhat variable in its effects. Where it was most severe, olivine is pseudomorphed by tremolite–actinolite, magnetite is replaced by a turbid mixture of sphene and undetermined "dust," and chromite is partly altered to a nonmetallic, semiopaque substance. If plagioclase was present it is now represented by albite, which pseudomorphs the original crystal, and by epidote, which crystallizes separately. However, generally some clinopyroxene survives and chromite is little altered. Only rarely is fresh olivine present but where the metamorphism was low chlorite commonly replaces olivine. On average, the temperature of metamorphism was probably not less than about 350°C (cf. Winkler 1976, p. 188).

In only a few samples are minerals identified that could be responsible for the magnetic signatures obtained in thermal demagnetization. In most of these the only magnetic mineral recognized is zoned chromite–magnetite. The stable magnetization with blocking temperatures near 580°C may be carried by the magnetite rim, whereas that with somewhat lower blocking temperatures may be carried by the chromite–magnetite solid solution. Even chromites that show no apparent zoning may contain sufficient iron at their margins as to be slightly magnetic up to moderate blocking temperatures. Note, in this regard, the Curie temperature curve of chromite–magnetite solid solution given by Bleil and Petersen (1982, p. 345) where the Curie temperature changes continuously between the end members. Any such solid solution with Cr/Fe atomic ratio less than about 1:2 would have a Curie temperature greater than 100°C and could contribute to the remanent magnetization of the specimen.

Microprobe analyses of examples of the cores and rims of zoned chromite crystals are given in Table 4. Note that the TiO_2 content of the magnetite rim is low in comparison with that of titanomagnetites generally found in mafic igneous rocks. This could account for its preservation under these metamorphic conditions if the texture were primary. Six of the seven examples that have distinctly zoned chromite–magnetite crystals also have high magnetic intensities and thermal demagnetization curves indicative of the presence of nearly pure magnetite (e.g., 2701 of Fig. 8*a*, 3203 of Fig. 9). The seventh has an intermediate intensity. Hence, in these samples it seems likely

TABLE 3. Petrography of Ottawa Islands paleomagnetic samples

Site	Core	Int.	MD	T_B range (°C)	Primary silicates	"Opaque" minerals		Remarks
25	06	I	A	400–560	Cpx 25%	Chromite Magnetite Altered prod.	1.7%	Finer chromite crystals thinly mantled with magnetite; no free magnetite; chromite 50% altered
26	01	I	B	175–350		Chromite Altered prod.	0.6%	
						Pyrite	Trace	
	04	I	A VRM	400–560 20–175	Cpx 25% Pl 15–20%	Ilmenite Altered prod.	2.7%	Magnetite pseudomorphs with ilmenite lamellae
						Sulphide	Rare	
27	01	H	B	100–560	Ol 5%	Chromite Magnetite Altered prod.	7.5%	Magnetite mantles chromite crystals; some semiopaque alteration products after olivine
						Pyrite	Rare	
30	04	H	A VRM	400–560 20–400	Cpx 25%	Chromite Magnetite Altered prod.	8.8%	Olivine pseudomorphs charged with semiopaque alteration products plus chromite crystals; some chromites mantled by thin magnetite rims
						Pyrite	Rare	
31	05	H	A VRM	400–550 20–250	Cpx 15%	Chromite Magnetite? Altered prod.	3.1%	Chromite typically with altered rims dominant; magnetite in semiopaque masses uncertain
						Chalcopyrite	Rare	
32	03	H	A and B VRM	450–560 20–450	Ol 15% Cpx 15% Pl 12%	Chromite Magnetite Altered prod.	1.7%	Relatively fresh rock; chromite with magnetite rims; no free magnetite; olivine charged with semiopaque alteration
						Chalcopyrite	Minor	
33	01	L	A VRM	300–540 20–175	Cpx Minor	Chalcopyrite Pyrite Chromite	Rare	Almost no opaques
						Altered prod.	Rare	
	03	L	B	250–350?	Cpx Minor	Pyrite Chalcopyrite	Rare	Almost no opaques
	04	L	A and B	300–500	Cpx 10–15%	Chalcopyrite Pyrite Chromite Altered prod.	Rare	Almost no opaques
34	02	H	A			Chromite Magnetite	<1%	Chromite with thin magnetite rims; dispersed granular sphene in groundmass after magnetite
						Chalcopyrite	Rare	
	05	L	A	250–530 Largely 300–350	Ol 30–40% Cpx 15%	Chromite Chalcopyrite	Rare	Spinifex-textured komatiite, fairly fresh; almost no opaques
35	04	L	A			Chalcopyrite Pyrite	Minor	Only recognizable opaques are sulphides confined to olivine pseudomorphs
	05	L	A B	400–530 250–350		Chalcopyrite Pyrite	Minor	Opaques minor; all show turbid, semiopaque alteration mantles
						Chromite	Rare	
						Altered prod.	Rare	
36	04	L	A B	400–500 175–300		Chromite	Rare	Almost no opaques
						Chalcopyrite Pyrite	1%	
37	02	I	A and B	250–500		Pyrite	Rare	Magnetite pseudomorphed by granular sphene shows pinpoint reflectivity
						Altered prod.	4%	
38	04	L	A and B			Chromite	Rare	Almost no opaques
						Pyrite Chalcopyrite	Minor	

TABLE 3. (*Concluded*)

Site	Core	Int.	MD	T_B range (°C)	Primary silicates	"Opaque" minerals		Remarks
39	02	H	A			Chromite Magnetite Pyrite Altered prod.	10%	Chromite mantled by magnetite rims and commonly turbid, semiopaque alteration product; majority of opaques in olivine pseudomorphs
40	04	L	A and B	20–530	Ol Rare Cpx 10%	Chromite Chalcopyrite Pyrite Altered prod.	Rare <1%	Almost no opaques
41	01	L	A and B?		Cpx 15%	Altered prod.	3%	Magnetite pseudomorphed by granular sphene with pinpoint reflectivity
42	03	I	A and B VRM	250–500 20–175	Cpx 10%	Chromite Chalcopyrite Pyrite Altered prod.	Rare Minor	Almost no opaques
44	02	H	A VRM	350–560 20–350	Ol 3% Cpx 20%	Chromite Magnetite? Altered prod. Chalcopyrite	4–5% Rare	Opaques are mainly in olivine pseudomorphs; chromite and magnetite? altered to turbid semiopaque; possibly thin magnetite rims on chromite
45	03	L	A	250–500		Pyrite Altered prod.	Rare Minor	Clots of turbid granular sphene probably after magnetite

NOTES: Int. is the NRM intensity (H = high; I = intermediate; L = low); MD is the direction of magnetization eliminated over a given blocking temperature (T_B) range; VRM is viscous remanent magnetization; Cpx is clinopyroxene; Pl is plagioclase; Ol is olivine.

TABLE 4. Cores and rims of zoned chromites from the Ottawa Islands and typical chromite from ultramafic rocks and magnetite from basalts

	2701		3402		Chromite[a] in ultramafic	Magnetite[b] in basalt
	Core	Rim	Core	Rim		
Cr_2O_3	46.96	7.46	48.08	39.40	51.54	0.19
Al_2O_3	13.00	1.23	11.32	9.75	13.67	2.38
FeO*	30.03	89.29	36.23	43.26	21.50	69.01
MgO	8.46	0.62	1.06	1.36	11.98	1.30
TiO_2	0.66	0.26	0.42	0.45	0.00	24.65
MnO	0.00	0.07	1.29	1.62	0.32	0.63
SiO_2	0.36	0.48	0.45	1.81	0.36	1.02
Na_2O	0.37	0.59	1.01	1.89	—	0.00
K_2O	0.00	0.00	0.00	0.02	—	0.00
CaO	0.16	0.07	0.21	0.43	—	0.00

NOTES: Microprobe analyses of chromites performed by M. Bonardi, Geological Survey of Canada; FeO* = total iron reported as FeO.

[a]Chromite from serpentines, Hiroosa mine, Japan. Analyst, J. Kitahara. Reported in Deer *et al.* (1962).

[b]Magnetite in Natkusiak basalt sample FX7541, Victoria Island, N.W.T. Analyst, M. Bonardi, Geological Survey of Canada.

that a major part of the remanent magnetization is carried by the chromite–magnetite zoned crystals. Zoned chromite crystals have been attributed to both magmatic (Frisch 1971; Haggerty 1976) and metamorphic processes (Bliss and MacLean 1975). However, zoning to nearly pure magnetite, as is the case in samples from the Ottawa Islands, was interpreted by Bliss and MacLean as the product of regional metamorphism at epidote amphibolite or higher facies levels. In two of the samples containing the zoned chromite–magnetites olivine pseudomorphs are also charged with magnetite and an accompanying turbid, semiopaque, but apparently nonmetallic, substance. This may be inherited from primary high-temperature oxidation of the olivine. If so, it could carry a primary magnetization direction.

For all other samples the magnetic carrier is not apparent in polished thin section and its identification must be speculative. Both the secondary sphene after magnetite and the opaque alteration substances that locally mantle chromite and sulphide minerals generally show poorly defined spot and pinpoint reflectivity in polished section that could be either relict primary or secondary magnetite and (or) ilmenite solid solution. Alternatively, submicroscopic magnetite particles that may have survived in some of the relict primary silicates would be capable of carrying a major part of the stable remanent magnetization (Hargraves and Young 1969). This is a factor that is difficult to assess. However, it should be noted that there is very little correlation between the proportion of primary silicates present in the rock and the intensity of its magnetization. Moreover, it is not obvious why such minute particles should survive when the major part of the coarser grained magnetite has not.

Origin of A and B magnetizations

The origin and relative age of the A and B components are critical to an evaluation of the paleomagnetic results from the Ottawa Islands. From directional data, only the A magnetization can be dated relative to folding. It is not immediately obvious whether the B component is pre- or postfolding, or whether it was acquired before or after the A magnetization. The nature of the A and B components is considered in detail below.

In cases of magnetic overprinting a simple examination of the blocking temperature spectra of the superimposed components is not sufficient to establish their relative age. This is obvious in the case of the Ottawa Islands where at least one of the magnetizations must be, in part, of chemical origin in order

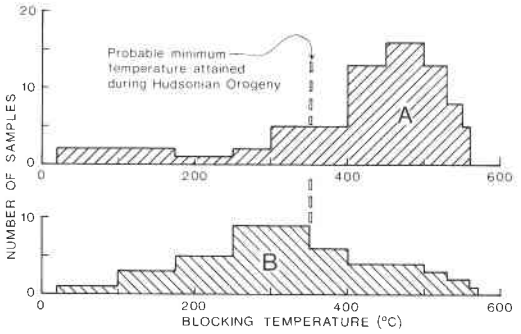


FIG. 13. Range of A and B blocking temperatures estimated from all samples used in thermal demagnetization experiments is compared with the probable minimum temperature attained during the Hudsonian Orogeny (estimated from Winkler 1976, p. 188).

to explain the large overlap of A and B blocking temperature spectra described above.

McClelland-Brown (1982) has determined the maximum overlap that would be observed at different temperatures if coexisting single-domain magnetite and hematite carry thermoremanent magnetizations. A greater overlap would indicate chemical remanent magnetization. These calculations are based on the variation in magnetic blocking temperature of each mineral with time (Pullaiah *et al.* 1975) or with cooling rate (Dodson and McClelland-Brown 1980).

Unfortunately, in the present study the magnetic carriers are not fully defined. A and B blocking temperatures (Fig. 13) for all specimens used in the thermal demagnetization experiments verify the presence of magnetite (see petrography) and also suggest that titanomagnetite and the chromite–magnetite solid solution may carry a portion of the remanence. There is no evidence for hematite. Thus the simple magnetite–hematite calculation of McClelland-Brown (1982) cannot be used. Nevertheless, the large overlap of A and B components undoubtedly indicates the presence of chemical remanent magnetization.

The postfold A magnetization of the Ottawa Islands has a paleopole similar to paleopoles interpreted as overprints related to the Hudsonian Orogeny (see Discussion) when temperatures in the Ottawa Islands probably reached at least 350°C. It is evident in Fig. 13 that A blocking temperatures are usually above 350°C and are frequently as high as 560°C. If the reheating temperature during the Hudsonian Orogeny did not greatly exceed 350°C, the A component is likely, at least in part, a chemical remanent magnetization rather than of thermal origin. This would indicate that the magnetite rims on chromite, which are often associated with the A component, are indeed of secondary origin.

The B magnetization is associated with minerals with a wide range of blocking temperatures (Fig. 13). Commonly blocking temperatures are 200–500°C. The portion of the B component carried by blocking temperatures near 200°C cannot have survived reheating to a minimum of 350°C during the Hudsonian Orogeny. Although it is possible that the temperature did not everywhere reach that indicated by the incompletely established secondary mineralogy, it seems unlikely that the B component could predate the Hudsonian Orogeny. If, as indicated above, the A component was acquired as an overprint related to the Hudsonian Orogeny, the B component is probably younger than the A. Since the upper (>400°C) portion of the B blocking temperature spectrum overlaps the A blocking temperature range (Fig. 13), this portion of the B magnetization must be of chemical origin, acquired at temperatures insufficient to thermally remagnetize the earlier A component.

Discussion

Paleopoles from the Circum-Superior Belt and Proterozoic outliers of the Superior Province (see sampling localities in Fig. 1) fall into three groups (Schwarz and Freda 1983).

Group 1 poles (Fig. 14*a*) include the prefold Eskimo volcanics pole (EV) of the Belcher Islands and several poles from Richmond Gulf, two of which (PN and PS of the Persillon Formation) are prefold.

Group 2 poles include five prefold results. The poles for the Flaherty volcanics (FV) and Haig intrusions (HI) of the Belcher Islands, the Sakami Formation (SAK) pole of central Quebec, and the Chakonipau Formation north (CHI) and south (CH2) poles from the Labrador Trough all belong to this grouping. On the basis of stratigraphic evidence from the Belcher Islands, the Eskimo volcanics are thought to be older than the Flaherty volcanics and Haig intrusives. Therefore, if the corresponding prefold paleopoles are primary, group 1 poles must be older than group 2 poles.

Group 3 poles, on the other hand, are postfold and hence younger than both group 1 and group 2 poles. Group 3 includes the postfold overprint of the Eskimo volcanics (EO) on the Belcher Islands, the postfold Cape Smith (CS) and Ottawa Islands A poles, and paleopoles of the Seward arkoses (SB) and Menihek argillites (MB) from a reconnaissance study of the Labrador Trough.

Group 3 paleopoles, including the Ottawa Islands A pole, fall at about 1700 Ma on the North American apparent polar wander path (e.g., dashed band of Fig. 14*a* follows Irving 1979) and undoubtedly represent magnetic overprinting related to the Hudsonian Orogeny. Group 1 and group 2 poles, however, do not fit as conveniently on this path.

Group 1 paleopoles have been used to define the depth of the Coronation Loop of the North American path (e.g., Schmidt 1980), as illustrated by the group 1 antipoles of Fig. 14*a*. Such an interpretation would require that group 1 poles be 1750–1800 Ma old. McGlynn and Irving (1981), however, have emphasized the uncertainty in extending the Coronation Loop through paleopoles that have not been corroborated from elsewhere in the Canadian Shield.

Further problems are encountered if group 2 paleopoles are assumed to fall on the North American polar wander path. Since they are likely younger than group 1 poles and older than group 3 poles, as discussed above, they would fall on the 1700 Ma portion of the path, just slightly older than the group 3 Hudsonian overprint poles (e.g., Schmidt 1980).

Schwarz and Freda (1984) suggested that group 1 and group 2 poles may define a path that roughly parallels the North American path between 2100 and 1800 Ma (solid band of Fig. 14*a*) and presumably joins the North American track before (or at the time of acquisition of) the Hudsonian overprint poles. They noted that the difference between the two paths may indicate relative movement between the Ungava Peninsula and the rest of the Canadian Shield during pre-Hudsonian times.

Although the A magnetization of the Ottawa Islands was undoubtedly acquired during uplift that followed the Hudsonian Orogeny, the origin of the B component is less certain. If primary, it should lie on the pre-Hudsonian portion of the polar wander path. However, the B paleopole (see estimated positions before and after tilt correction in Fig. 14*a*) does not

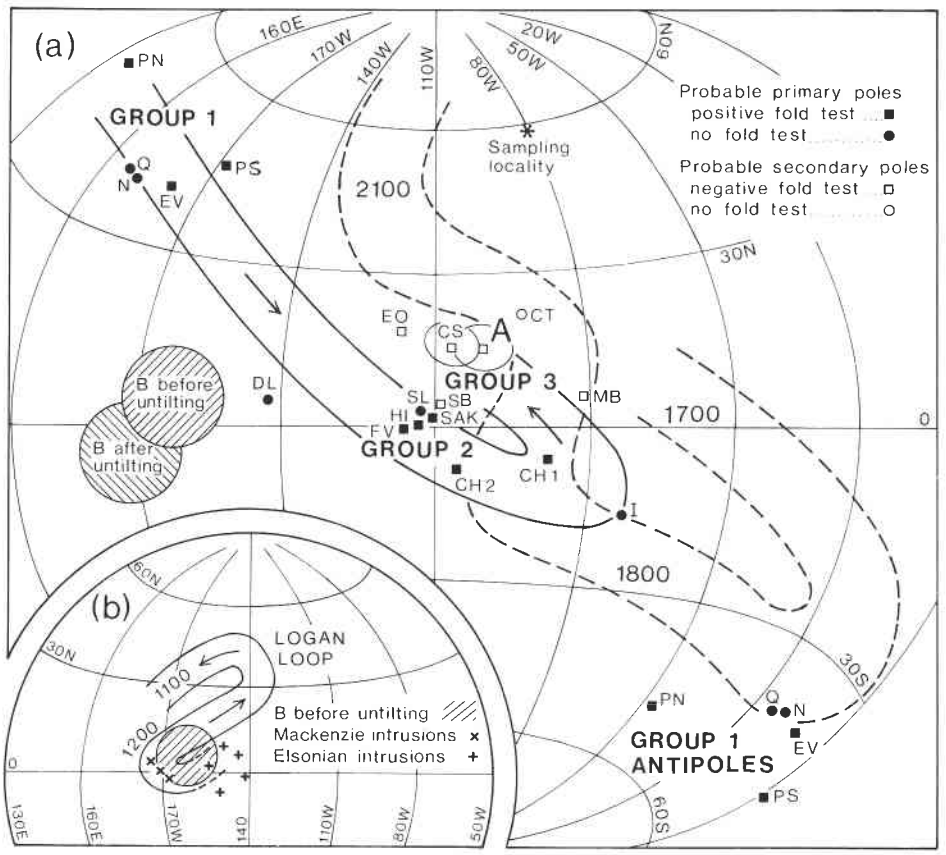


FIG. 14. Ottawa Islands paleopoles and the apparent polar wander path for Precambrian North America. The shaded circles give only an estimate of the B pole location and do not indicate a pole and its circle of confidence as defined by Fisher statistics. (*a*) The Ottawa Islands A paleopole before untilting and its circle of 95% confidence and estimates for the B pole before and after untilting are compared with the North American polar wander path of Irving (1979) between 2100 and 1700 Ma and the Circum-Superior path of Schwarz and Freda (1983) for the pre-1700 Ma period (dashed and solid curves, respectively). Paleopoles from the Circum-Superior Belt are (A), (B) Ottawa Islands; (CH1), (CH2) Chakonipau Formation; (CS) Cape Smith volcanics; (CT) Castignon Complex; (DL) Dieter Lake redbeds; (EV) Eskimo volcanics; (EO) Eskimo overprint; (FV) Flaherty volcanics; (HI) Haig intrusions; (I) Lac Imbault redbeds; (MB) Menihek Formation; (N) Nastapoka basalts; (PN) Persillon Formation (north); (PS) Persillon Formation (south); (Q) Qingaaluk sediments; (SAK) Sakami Formation; (SB) Seward Subgroup; (SL) Sulton Lake inlier. (*b*) The estimate of the B paleopole before untilting is compared with the 1200–1100 Ma Logan Loop (after Berger and York 1980) and poles of the 1250 Ma Mackenzie igneous episode and the 1450 Ma Elsonian plutons.

fit the pre-Hudsonian portion of the polar wander path in either of the above interpretations. Rather it lies well to the west of group 2 poles. A substantial alteration of the path suggested by Schwarz and Freda (1983) would be required to incorporate the B pole.

As suggested in the earlier discussion of the origin of the magnetic components of the Ottawa Islands, however, the B remanence is most likely secondary and, in fact, probably postdates the A magnetization. A comparison of the estimated location of the B paleopole with the post-Hudsonian polar wander path (Fig. 14*b*) indicates that the B magnetization predates the 1200–1100 Ma Logan Loop. The B pole falls near poles of the 1250 Ma Mackenzie igneous episode and the 1450 Ma Elsonian plutons, which are both well documented tie points on the path. Indeed, several K–Ar and one Rb–Sr data from the Cape Smith Belt fall in the range 1430–1550 Ma, suggesting that the B magnetization may have been acquired during a period of mild reheating associated with the Elsonian disturbance in Labrador.

It is interesting to note that a direction somewhat similar to the antipole of the B magnetization was reported (Schwarz and Fujiwara 1981) at several sites in the Manitounuk Islands, which lie along the east coast of Hudson Bay south of Richmond Gulf. A shallow, easterly magnetization was obtained at six sites, and a magnetization similar to the Ottawa Islands A component was observed at two others. No paleopole was calculated, since the authors were uncertain (a) that secular variation had been averaged out, and (b) that only a single component of magnetization was involved. The relative age of the components in the Manitounuk Islands was not discussed. Nevertheless, the similarity with the B direction of the Ottawa Islands may imply that the post-Hudsonian overprinting of the Circum-Superior belt of eastern Hudson Bay was not confined to the Ottawa Islands.

Conclusions

The komatiitic basalts of the Ottawa Islands carry two stable components of remanent magnetization. The dominant A component is a postfold remanence probably acquired as an overprint related to the Hudsonian Orogeny. The B component, which is less well defined, is most easily explained as a later overprint possibly acquired at the time of the Elsonian disturbance in Labrador.

Acknowledgments

We wish to thank K. R. Clark and C. G. Lamontagne, who

carried out the paleomagnetic sampling of the Ottawa Islands. Comments by D. J. Dunlop on the nature of stable and swinging intermediate A–B directions and by P. W. Schmidt on the significance of overlapping blocking temperatures spectra were appreciated.

ARNDT, N., and BROOKS, C. 1980. Komatiites. Geology, **8**, pp. 155–156.

BAILEY, M. E., and HALE, C. J. 1981. Anomalous magnetic directions recorded by laboratory-induced chemical remanent magnetization. Nature, **294**, pp. 739–741.

BARAGAR, W. R. A. 1974. Volcanic studies in the Cape Smith – Wakeham Bay Belt, New Quebec. *In* Report of activities, part A. Geological Survey of Canada, Paper 74-1A, pp. 155–157.

——— 1983. The Circum-Ungava Belt of eastern Hudson Bay: geology of the Cape Smith region. *In* Current research, part A. Geological Survey of Canada, Paper 83-1A, pp. 325–328.

BARAGAR, W. R. A., and LAMONTAGNE, C. G. 1980. The Circum-Ungava Belt in eastern Hudson Bay: the geology of Sleeper Islands and parts of the Ottawa and Belcher islands. *In* Current research, part A. Geological Survey of Canada, Paper 80-1A, pp. 89–94.

BARAGAR, W. R. A., and PICHÉ, M. 1982. The Circum-Ungava Belt of eastern Hudson Bay: the geology of the Ottawa Islands and Cape Smith region. *In* Current research, part A. Geological Survey of Canada, Paper 82-1A, pp. 11–15.

BEALL, G. H., HURLEY, P. M., FAIRBAIRN, H. W., and PINSON, W. H. 1963. Comparison of K–Ar and whole rock Rb–Sr dating in New Quebec and Labrador. American Journal of Science, **261**, pp. 571–580.

BERGER, G. W., and YORK, D. 1980. Reinterpretation of the North American apparent polar wander curve for the interval 800–1500 Ma. Canadian Journal of Earth Sciences, **17**, pp. 1229–1235.

BERGERON, R. 1959. Preliminary report on Povungnituk Range area, New Quebec. Quebec Department of Mines, P.R. 392, 9 p.

BLEIL, U., and PETERSEN, N. 1982. Magnetic properties of minerals. *In* Numerical data and functional relationships in science and technology. Vol. 1. Physical properties of rocks. *Edited by* G. Angenheister. Springer-Verlag, Berlin. pp. 308–365.

BLISS, N. W., and MACLEAN, W. H. 1975. The paragenesis of zoned chromite from central Manitoba. Geochimica et Cosmochimica Acta, **39**, pp. 973–990.

DEER, W. A., HOWIE, R.A., and ZUSSMAN, J. 1962. Rock forming minerals. Vol. 5. Longmans, Green, and Co., London, 371 p.

DODSON, M. H., and MCCLELLAND-BROWN, E.A. 1980. Magnetic blocking temperatures of single-domain grains during slow cooling. Journal of Geophysical Research, **85**, pp. 2625–2637.

FRANCIS, D. M., and HYNES, A. J. 1979. Komatiite-derived tholeiites in the Proterozoic of New Quebec. Earth and Planetary Science Letters, **44**, pp. 473–481.

FRISCH, T. 1971. Alteration of chrome spinel in a dunite nodule from Lanzarote, Canary Islands. Lithos, **4**, pp. 83–91.

FRYER, B. J. 1972. Age determinations in the Circum-Ungava geosyncline and the evolution of Precambrian banded iron formations. Canadian Journal of Earth Sciences, **9**, pp. 652–664.

FUJIWARA, Y., and SCHWARZ, E. J. 1975. Paleomagnetism of the Circum-Ungava Proterozoic Fold Belt (I): Cape Smith komatiitic basalts. Canadian Journal of Earth Sciences, **12**, pp. 1785–1793.

HAGGERTY, S. E. 1976. Oxidation of opaque mineral oxides in basalts. *In* Oxide minerals. *Edited by* D. Rumble. Mineralogical Society of America, Short Course Notes, pp. Hg-1–Hg-100.

HALLS, H. C. 1979. Separation of multicomponent NRM: combined use of difference and resultant magnetization vectors. Earth and Planetary Science Letters, **43**, pp. 303–308.

HARGRAVES, R. B., and YOUNG, W. M. 1969. Source of stable remanent magnetism in Lambertville diabase. American Journal of Science, **267**, pp. 1161–1177.

HOFFMAN, K. A., and DAY, R. 1978. Separation of multi-component NRM: a general method. Earth and Planetary Science Letters, **40**, pp. 433–438.

IRVING, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**, pp. 669–694.

MCCLELLAND-BROWN, E. 1982. Discrimination of TRM and CRM by blocking-temperature spectrum analysis. Physics of the Earth and Planetary Interiors, **30**, pp. 405–414.

MCGLYNN, J. C., and IRVING, E. 1981. Horizontal motions and rotations in the Canadian Shield during the Early Proterozoic. *In* Proterozoic basins of Canada. *Edited by* F. H. A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 183–190.

PARK, J. K. 1977. A reconnaissance paleomagnetic survey of the central Labrador Trough, Quebec. Canadian Journal of Earth Sciences, **14**, pp. 159–174.

PULLAIAH, G., IRVING, E., BUCHAN, K. L., and DUNLOP, D. J. 1975. Magnetization changes caused by burial and uplift. Earth and Planetary Science Letters, **28**, pp. 133–143.

SCHIMANN, K. 1978. Geology of the Wakeham area, eastern end of the Cape Smith Belt, New Quebec. Ph.D. thesis, University of Alberta, Edmonton, Alta., 426 p.

SCHMIDT, P. W. 1980. Paleomagnetism of igneous rocks from the Belcher Islands, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **17**, pp. 807–822.

SCHWARZ, E. J., and FREDA, G. N. 1983. Paleomagnetism and time-stratigraphic correlation of Proterozoic redbeds of the Labrador Trough and outliers of northern Quebec. Canadian Journal of Earth Sciences, **20**, pp. 1725–1737.

SCHWARZ, E. J., and FUJIWARA, Y. 1977. Komatiitic basalts from the Proterozoic Cape Smith range in northern Quebec, Canada. Geological Association of Canada, Special Paper 16, pp. 193–201.

——— 1981. Paleomagnetism of the Circum-Ungava Fold Belt II: Proterozoic rocks of Richmond Gulf and Manitounuk Islands. *In* Proterozoic basins of Canada. *Edited by* F. H. A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 255–267.

SCHWARZ, E. J., CLARK, K. R., and FUJIWARA, Y. 1982. Paleomagnetism of the Sutton Lake Proterozoic inlier, Ontario, Canada. Canadian Journal of Earth Sciences, **19**, pp. 1330–1332.

SEGUIN, M. K., SHARMA, K. N. M., and WOUSSEN, G. 1981. Étude paléomagnétique des roches protérozoïques de la formation de Sakami, région de la Grande Rivière, Territoire du Nouveau-Québec, Canada. Canadian Journal of Earth Sciences, **18**, pp. 1893–1899.

STEARN, J. E. F., and PIPER, J. D. A. 1984. Paleomagnetism of the Sveconorwegian Mobile Belt of the Fennoscandian Shield. Precambrian Research, **23**, pp. 201–246.

STOCKWELL, C. H. 1982. Proposals for time classifications and correlation of Precambrian rocks and events in Canada and adjacent areas of the Canadian Shield. Part 1: A time classification of Precambrian rocks and events. Geological Survey of Canada, Paper 80-19, 135 p.

TAYLOR, F. C. 1974. Reconnaissance geology of a part of the Precambrian Shield, northern Quebec and Northwest Territories. Geological Survey of Canada, Paper 74-21.

——— 1982. Reconnaissance geology of a part of the Canadian Shield, northern Quebec and Northwest Territories. Geological Survey of Canada, Memoir 399, 32 p.

WINKLER, H. G. F. 1976. Petrogenesis of metamorphic rocks. 4th ed. Springer-Verlag, New York, 334 p.

ZINDLER, A. 1981. Nd and Sr isotopic studies of komatiites and related rocks. *In* Komatiites. *Edited by* N. Arndt and E. G. Nisbet. Allen & Unwin Inc., Winchester, MA, 526 p.